Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)

Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

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1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

2. Supervisor: Dr. Name Surname
Name of Research Centre or Department
University *of* Limerick
Ireland

1. Reviewer: Dr. Name Surname
Name of Research Centre or Department
University of
Country

2. Reviewer: Dr. Name Surname
Name of Research Centre or Department
University of
Country

1. Chair: Dr. Name Surname
Name of Research Centre or Department
University *of* Limerick
Ireland

Day of the defense: day month year

Signature from head of PhD committee:

Abstract

Falls, frailty, and chronic conditions such as cardiovascular disease, diabetes, and obesity remain leading causes of morbidity and mortality in older adults. Yet, existing predictive models often rely narrowly on physical health indicators, overlooking nutrition and lifestyle factors that may enhance early detection of risk. This thesis develops and evaluates machine learning models using data from The Irish Longitudinal Study on Ageing (TILDA) to predict key health outcomes in a cohort of older adults. The models integrate physical, mental, lifestyle, and socio-demographic predictors, with particular emphasis on under-explored nutritional and behavioural variables highlighted in prior research. Multiple approaches are compared, including logistic regression, random forests, and gradient boosting, with performance assessed via cross-validation and feature importance analysis. By identifying high-impact, modifiable predictors, this work aims to advance early risk stratification, guide targeted prevention strategies, and inform public health policy to support healthy ageing.

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

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Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

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List of Abbreviations

AUC Area Under the ROC Curve

CES-D Center for Epidemiologic Studies Depression Scale

DNN Deep Neural Network

FI Frailty Index

GBM Gradient Boosting Machine
GDS Geriatric Depression Scale

HRS Health and Retirement Study, (a longitudinal study of ageing in the United States)

LDA Linear Discriminant Analysis

ML Machine Learning
MPOD Macular pigment optical density

RF Random Forest

SHAP Shapley Additive Explanations

TILDA The Irish Longitudinal Study on Ageing
TUG Timed Up and Go (mobility test)

WHO World Health Organization

List of Abbreviations

Chapter 1

Introduction

1.1 Background and Context

Chronic conditions—including cardiovascular disease, diabetes, and frailty—are leading contributors to morbidity and mortality in ageing populations [1, 2]. These conditions are associated with declines in functional capacity (e.g., gait speed, grip strength [3]), increased healthcare utilisation [4], and reduced quality of life [5].

Traditional risk prediction models rely primarily on clinical biomarkers such as blood pressure and lipid profiles. Evidence suggests that multidimensional predictors, including nutrition [6], physical activity [7], mental health [8], and social determinants [9], improve predictive performance.

The Irish Longitudinal Study on Ageing (The Irish Longitudinal Study on Ageing (TILDA)) provides a nationally representative dataset of adults aged 50+ in Ireland with repeated measures across multiple domains. The Harmonized TILDA dataset aligns variables with the RAND Health and Retirement Study (Health and Retirement Study, (a longitudinal study of ageing in the United States) (HRS)) framework for cross-country comparability, while the raw waves contain additional study-specific variables. The IDS dataset supplements coverage for participants with disabilities.

These data motivate the development of integrated models combining traditional (e.g., blood pressure, cholesterol, diabetes diagnosis) and non-traditional (e.g., nutrition biomarkers, physical activity, mental health, social determinants) risk factors. This thesis focuses on the non-traditional predictors to quantify their added predictive value.

1.2 Research Problem and Motivation

Existing Machine Learning (ML) approaches for chronic disease prediction in older adults face several challenges:

1.2.1 Limited Integration of Multidimensional Data

Most models prioritise clinical variables but neglect modifiable lifestyle and nutrition factors [6, 10].

1.2.2 Underuse of Longitudinal Dynamics

Few studies exploit repeated measures to capture trajectories of risk, such as frailty progression [11].

1.2.3 Lack of Explainability

Many ML models are not interpretable, limiting clinical utility [12].

Addressing these gaps requires methods that integrate multidomain predictors, account for longitudinal dynamics, and balance predictive accuracy with interpretability.

1.3 Research Aims and Objectives

This thesis develops and validates ML models that integrate nutrition, lifestyle, and mental health features with clinical data to improve prediction of chronic disease and ageing-related outcomes.

- Conduct a literature review on ML for chronic disease prediction in ageing populations, emphasising gaps in lifestyle and nutrition integration.

- Preprocess and harmonise TILDA waves 1–5 and engineer multidomain features (nutrition, activity, mobility, mental health, clinical indicators).
- Develop and compare baseline statistical models (logistic regression) with advanced ML methods (Random Forest (RF), Gradient Boosting Machine (GBM)).
- Evaluate models using discrimination (Area Under the ROC Curve (AUC)), calibration, and error metrics (accuracy, precision, recall, F1-score); perform ablation analyses to quantify added value of lifestyle and nutrition.
- Identify actionable predictors (e.g., vitamin D status, gait speed reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.4 Methodology Overview

This section summarises the workflow; detailed methods are presented in Chapter 3.

Data are derived from both raw TILDA waves 1–5 and the Harmonized TILDA dataset, aligned to the HRS framework. IDS data complements coverage for participants with disabilities.

Missing data are addressed with complete-case analysis and multiple imputation. Derived features capture multidimensional predictors:

- Frailty indices (e.g., Fried's phenotype, Frailty Index (FI))
- Mobility measures (e.g., Timed Up and Go (mobility test) (TUG), gait speed, grip strength)
- Mental health scales (e.g., Center for Epidemiologic Studies Depression Scale (CES-D), Geriatric Depression Scale (GDS))
- Nutrition scores (e.g., dietary diversity indices, micronutrient biomarkers)

Baseline (logistic regression) and advanced ML models (RF, GBM) are tuned via cross-validation. Performance is evaluated using discrimination (AUC), calibration (plots, Brier score), and error metrics. Explainability is assessed with feature importance and Shapley Additive Explanations (SHAP) values.

1.5 Contributions of the Research

- **Empirical:** Quantifies the predictive value of nutrition and lifestyle features beyond clinical baselines.
- **Methodological:** Provides a reproducible pipeline for harmonising longitudinal TILDA data and integrating multidomain features into ML models.
- **Clinical:** Highlights modifiable predictors for targeted interventions and policy to promote healthy ageing.

1.6 Thesis Outline

- *Chapter Two:* Literature review on ML prediction of chronic disease in ageing populations.
- *Chapter Three:* TILDA study design, data preprocessing, harmonisation, and feature engineering.
- *Chapter Four:* Modelling framework, including baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics.
- *Chapter Five:* Results, model comparison, feature importance, and ablation analyses.
- *Chapter Six:* Discussion of implications, limitations, ethical considerations, and future work.

A series of documents have been included in the Appendix section of this dissertation. These are:

- *Appendix A*: Variable codebook and wave harmonisation notes (including anonymisation actions, top-coding, and grouping rules relevant to the features used).
- *Appendix B*: Ethics.
- *Appendix C*: Modelling details, additional tables/figures, and reproducibility checklist (overview of data processing scripts).

A digital archive is attached to this dissertation containing:(Access request required)

- *folder 1*: Preprocessing and feature-engineering scripts (documentation only; no raw TILDA data).
- *folder 2*: Model training/evaluation code and configuration files to reproduce reported results.

Chapter 2

Literature Review

2.1 Introduction

- Purpose of this chapter
- Brief overview on TILDA as a unique dataset (rich, longitudinal, multi-domai...)
- Scope: frailty, mental health, chronic disease, multimorbidity, nutrition, and ML models

2.2 Chronic Disease and Ageing

- The burden of chronic conditions in older adults
- Healthy ageing framework World Health Organization (WHO)
- Importance of multimorbidity, frailty, mental health

2.2.1 Frailty and Physical Function

- Gait speed, grip strength, functional mobility decline
- Predictors of frailty transitions ([13], [14], [15])

2.2.2 Mental Health in Older Adults

- Depression, anxiety, loneliness, social connectedness ([16], [17], [18])
- Links between physical activity and depression ([8], [7], [10])

2.2.3 Nutrition and Biological Ageing

- Vitamin D, lutein, zeaxanthin, Macular pigment optical density (MPOD) ([6], [19], lairdVitaminDeficiencyIreland2020, feeneyLowMacularPigment2013)
- Malnutrition and its predictors ([2])
- Biological vs chronological ageing ([20], [21])

2.3 Machine Learning in Ageing and Chronic Disease

- Overview: Why ML is valuable in ageing research [22]
- Transition from regression-based models → ensemble ML → deep learning(cite a example here)

2.3.1 Feature Sets in ML Studies

- Clinical, sociodemographic, lifestyle, cognition, psychosocial
- Over-reliance on physical health predictors vs underuse of nutrition/social variables

2.3.2 ML Models Used

- Logistic regression, Markov models, Linear Discriminant Analysis (LDA) (traditional)
- Ensembles RF, GBM, Deep Neural Network (DNN) – e.g., ([23]) diabetes prediction

- Unsupervised clustering for dementia ([24])
- Brain age prediction models ([25], [26])

2.3.3 Performance and Validation

- AUC, R^2 ranges across tasks (diabetes, gait speed, mortality, depression)
- Limitations: internal validation only, cross-sectional data, no generalisation

2.3.4 Explainability in ML for Ageing

- SHAP, TreeExplainer examples ([23], [27], [3])
- Challenges of black-box models in clinical practice

2.4 Gaps in the Literature

- Limited population diversity (mostly Irish, 50+)
- Over-reliance on physical predictors; neglect of nutrition, social, religious/spiritual, environmental variables
- Dominance of regression models; limited advanced ML such as neural networks, transformers
- Validation gaps – very few external validations, especially for ensemble and brain-age models
- Underexplored outcomes: pain, sleep, QoL
- Cross-sectional bias; weak time-series modelling
- Poor transparency on missing data handling
- Limited integration of nutrition biomarkers
- Lack of biological age prediction vs chronological
- Interpretability challenges in ML

2.5 Chapter Summary

- I need to recap the main findings from LR systematic mapping excel file
- Identify your contribution niche:
  - integrating nutrition + lifestyle + clinical predictors
  - ML + explainability
  - time-based prediction (longitudinal TILDA data)
- Lead into Methodology (Chapter 3)

Chapter 3

Data and Preprocessing

3.1 Insert a Figure

We refer to a figure in the text like this: (Figure 3.1).

Figure 3.1: Title of Figure - Description. [Source: (**?**)]

3.2 Creating a list

- **(2000)** item 1.
- **(2004)** item 2.
- **(2010)** item 3.
- **(2013)** item 4.

3.2.1 Creating a Table

Table 3.1: Table Title

	Resolution	Min	Max
Gyroscope	4.5mV / °/s	0.27 °/s	406 °/s
Accelerometer	600mV /g	0.002g	2g
Magnetometer	385mV/ gauss	0.317 gauss	6 gauss

3.3 Quote

An in-text quote is declared in the following way:

> Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! Recycling is the way forward! (CommonSense, p. 0)

Another way of doing it is:

There are in our existence spots of time,
That with distinct pre-eminence retain
A renovating virtue, whence-depressed
By false opinion and contentious thought,
Or aught of heavier or more deadly weight,
In trivial occupations, and the round
Of ordinary intercourse-our minds
Are nourished and invisibly repaired;
A virtue, by which pleasure is enhanced,
That penetrates, enables us to mount,
When high, more high, and lifts us up when fallen.
(**?** , verses 208-218)

Chapter 4

Methodology

4.1 Math

In-text Math

- A vector vector $\hat{A}_{ba}\{x_{ba}, y_{ba}, z_{ba}\}$
- Another vector $R^b\{\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}\}$ as:

A series of Equations:

$$\hat{t}_{1b} = \hat{A}_{ba} \tag{4.1}$$

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.2}$$

$$\hat{t}_{3b} = \hat{t}_{1b} \times \hat{t}_{2b} \tag{4.3}$$

$$R^a = R^b(R^i)^T = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}][\hat{t}_{1i}, \hat{t}_{2i}, \hat{t}_{3i}]^T \tag{4.4}$$

where the symbol ^{T} denotes transposition.

$$\hat{t}_{2b} = \frac{\hat{A}_{ba} \times \hat{M}_{bm}}{\mid \hat{A}_{ba} \times \hat{M}_{bm} \mid} \tag{4.5}$$

$$= \frac{(y_{ba}z_{bm} - y_{bm}z_{ba}, z_{ba}x_{bm} - z_{bm}x_{ba}, x_{ba}y_{bm} - x_{bm}y_{ba})}{\mid A_{ba} \mid\mid M_{bm} \mid \sqrt{1 - (\hat{A}_{ba} \cdot \hat{M}_{bm})^2}}$$

$$= \frac{-0.2338, -0.0931, -0.0651}{0.2601}$$

$$= (-0.8989, -0.3579, -0.2503)$$

which if normalised gives:

$$= (-0.8995, -0.3581, -0.2504)$$

A matrix

$$R^b = [\hat{t}_{1b}, \hat{t}_{2b}, \hat{t}_{3b}] = \begin{pmatrix} -0.2698 & -0.8995 & 0.3439 \\ 0.0035 & -0.3581 & -0.9337 \\ 0.9629 & -0.2504 & 0.0997 \end{pmatrix}$$

Chapter 5

Results

5.1 References

Open the file 'references_myphd' with BibTex. The file is located in '/7_references/'

The file contains a series of templates for the proper formatting of references according to the Harvard referencing style. Click on each reference in the text below to see how Latex has formatted the reference on the References section.

- **Book**: (**?** , p. 10) (page number required only if referring to an in-text quote)
- **Book Chapter**: (**?**)
- **Book Contribution**: (**?**)
- **MUSIC CD/DVD**: (**?**)
- **Journal Article**: (**?**)
- **Patent**: (**?**)
- **PhD/Master Dissertation**: (**?**)
- **Conference Paper**: (**?**)
- **Technical Report**: (**?**)

- **Unpublished Report**: (**?**)
- **Web Videos**(**?**) (. . . not the name of the person who uploaded the video though)
- **Web Paper**: (**?**)
- **Web Journal Article**: (**?**)
- **Webpage**: (**?**)
- **Downloaded Images or Sounds**: (**?**)

You can also you this kind of referencing if needed in the text:
Surname [**?**] said that . . .

Chapter 6

Conclusions and Future Directions

6.1 Summary

6.2 Conclusions

6.3 Contributions

6.4 Future Work

Appendix A

Insert a figure

Appendix

Appendix B

Title

Type here you text as usual . . .

Appendix C

Code for estimating attitude

AHRS_TRIAD.C

```
/*
 *
 *   Copyright (c) 2013, Giuseppe Torre
 *   All rights reserved.
 *
 *   Redistribution and use in source and binary forms, with or without
 *   modification, are permitted provided that the following conditions are met:
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 *   ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
 *   (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
 *   SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
 *
 */



#include "ext.h"
#include "ext_mess.h"
#include <string.h>
#include <stdio.h>
#include <math.h>
#define EPSILON 1e-6

//******************CALIBRATION STUFF*************************//
/* INTRODUCE THE OFFSET VALUES IN THE FOLLOWING VARIABLES IN ADC values */
#define ACCELX_OFFSET 2043 //initial offset in accelerometer X
#define ACCELY_OFFSET 2053 // initial offset in accelerometer Y
#define ACCELZ_OFFSET 2264//initial offset in accelerometer Z

#define MagX_OFFSET 1917 //initial offset in Magnetometer X
```

Appendix

```c
#define MagY_OFFSET 1813 // initial offset in Magnetometer Y
#define MagZ_OFFSET 1667//initial offset in Magnetometer Z
/* WE SHOULD CALIBRATE THE FOLLOWING VALUES FOR EACH PARTICULAR IMU axis */
#define ACCELX_Resolution 536
#define ACCELY_Resolution 661
#define ACCELZ_Resolution 559
#define MagX_Resolution 186
#define MagY_Resolution 189
#define MagZ_Resolution 170
//****************** END CALIBRATION STUFF   ************************//


void *this_class; // Required. Global pointing to this class

typedef struct _triad // Data structure for this object
{
        t_object m_ob;    // Must always be the first field; used by Max

        Atom m_args[9];   // we want our inlet to be receiving a list of 10 elements

        long m_value;     //inlet

        void *m_R1;             //these are all the outlets for the 3 X 3 Matrix
        void *m_R2;             // R1 --> firts top left cell ..R2 middle cell of the first
            row ...and so on
        void *m_R3;                                    //
        void *m_R4;                                    // End Outlets
            ________________________________________
} t_triad;

                        void *triad_new(long value);


                        void triad_assist(t_triad *triad, void *b, long msg, long arg, char *
                            s);
                        void triad_free(t_triad *triad);



                        void triad_list(t_triad *x, Symbol *s, short argc, t_atom *argv);




                        void MatrixByMatrix(double *Result,double *MatrixLeft,double *
                            MatrixRight);

                        void Matrix2Quat(double *Quat,double *Matrix);
                        void Quat2Matrix(double *Matrix, double *Quat);
                        void inverseQuat(double *InvQuat, double *RegQuat);
                        void NormQuat(double *YesQuat, double *NotQuat);
                        void Slerp(double *NewQuat, double *OldQuat, double *CurrentQuat);
                        void NormVect(double *YesVect, double *NotVect);
                        double orientationMatrix[9];
                        double Result[9];
                        int    i;
                        double InvorientationMatrix[9];

                        double temp[6];
                        double ref[6];
                        double vectAx[3];
                        double vectAy[3];
                        double vectAz[3];
                        double vectBx[3];
                        double vectBy[3];
                        double vectBz[3];
                        double MagnCrosProd_A;
                        double MagnCrosProd_B;
                        double accnorm, magnorm, earthnorm, VectAynorm, VectAznorm,
                            VectBynorm, VectBznorm;
```

```
                              double m[9], n[9];
                              double quat_e[4];
                              double invquat_e[4];
                              double mult;
                              double quat_new[4];
                              double quat_old[4];
                              double ecs,accex_ADCnumber,y,accey_ADCnumber,z,accez_ADCnumber,mx,
                                   magnx_ADCnumber,my,magny_ADCnumber,mz,magnz_ADCnumber;
// SLERP Variables
                              double trace, Suca;
                              double tol[4], omega, sinom, cosom, scale0, scale1, tez,
                                   orientationMatrixA[9];

int main(void)
{
        // set up our class: create a class definition
        setup((t_messlist**) &this_class, (method)triad_new, (method)triad_free, (short)
             sizeof(t_triad), 0L,A_GIMME, 0);
    addmess((method)triad_list, "list", A_GIMME, 0);
        addmess((method)triad_assist, "assist", A_CANT, 0);
        finder_addclass("Maths","triad");
        post("....I'm-TRIAD-Object!....from-AHRS-Library...",0);
        return 0;
}

/* ——————————triad_new ——————————*/

void *triad_new(long value)
{
        t_triad *triad;
        triad = (t_triad *)newobject(this_class);        // create the new instance and return
             a pointer to it
        triad->m_R4 = floatout(triad);
        triad->m_R3 = floatout(triad);
        triad->m_R2 = floatout(triad);
        triad->m_R1 = floatout(triad);

return(triad);
}

etc.

etc.
```

Appendix

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